

March 18, 2021

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50281333

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 04, 2021. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Green Bay
- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids
Pace Project No.: 50281333

Pace Analytical Services Green Bay

1241 Bellevue Street, Green Bay, WI 54302
Florida/NELAP Certification #: E87948
Illinois Certification #: 200050
Kentucky UST Certification #: 82
Louisiana Certification #: 04168
Minnesota Certification #: 055-999-334
New York Certification #: 12064
North Dakota Certification #: R-150

Virginia VELAP ID: 460263
South Carolina Certification #: 83006001
Texas Certification #: T104704529-14-1
Wisconsin Certification #: 405132750
Wisconsin DATCP Certification #: 105-444
USDA Soil Permit #: P330-16-00157
Federal Fish & Wildlife Permit #: LE51774A-0

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 50281333

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50281333001	Digester #4	Solid	03/03/21 13:45	03/04/21 12:45

REPORT OF LABORATORY ANALYSIS

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50281333

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50281333001	Digester #4	EPA 8081	KAV	21	PASI-I
		EPA 8081	KAV	8	PASI-I
		EPA 8082	RID	8	PASI-I
		EPA 8151A	BJW	3	PASI-I
		EPA 6010	KJE	6	PASI-I
		EPA 6010	KJE	8	PASI-I
		EPA 6020	DMT	10	PASI-I
		EPA 7470	ILP	1	PASI-I
		EPA 7471	ILP	1	PASI-I
		EPA 8270	JCM	67	PASI-I
		EPA 8270	JCM	18	PASI-I
		EPA 8260	TMW	55	PASI-I
		EPA 5030/8260	TMW	13	PASI-I
		SM 2540G	OAS	1	PASI-I
		SM 2540G	OAS	1	PASI-I
		SM 2540G	OAS	1	PASI-I
		EPA 9038	SWJ	1	PASI-I
		EPA 9045	HCF	1	PASI-I
		EPA 351.2	TMK	1	PASI-G
		EPA 353.2	SLB	2	PASI-I
		SM 4500-NH3 G	MMS	1	PASI-I
		SM 4500-CI-E	SLB	1	PASI-I

PASI-G = Pace Analytical Services - Green Bay
PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids								
Analytical Method: EPA 8081 Preparation Method: EPA 3546 Pace Analytical Services - Indianapolis								
Aldrin	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	309-00-2	
alpha-BHC	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	319-84-6	
beta-BHC	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	319-85-7	
delta-BHC	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	319-86-8	
gamma-BHC (Lindane)	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	58-89-9	
Chlordane (Technical)	<7880	ug/kg	7880	1	03/12/21 10:57	03/15/21 14:59	57-74-9	
alpha-Chlordane	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	5103-71-9	
gamma-Chlordane	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	5103-74-2	
4,4'-DDD	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	72-54-8	
4,4'-DDE	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	72-55-9	
4,4'-DDT	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	50-29-3	
Dieldrin	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	60-57-1	
Endosulfan I	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	959-98-8	
Endosulfan II	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	33213-65-9	
Endosulfan sulfate	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	1031-07-8	
Endrin	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	72-20-8	
Endrin aldehyde	<788	ug/kg	788	1	03/12/21 10:57	03/15/21 14:59	7421-93-4	
Heptachlor	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	76-44-8	
Heptachlor epoxide	<394	ug/kg	394	1	03/12/21 10:57	03/15/21 14:59	1024-57-3	
Toxaphene	<7880	ug/kg	7880	1	03/12/21 10:57	03/15/21 14:59	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	63	%.	10-129	1	03/12/21 10:57	03/15/21 14:59	2051-24-3	

8081 GCS Pest RV, TCLP

Analytical Method: EPA 8081 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 03/10/21 14:00 Initial pH: 8.47; Final pH: 5.16

Pace Analytical Services - Indianapolis

gamma-BHC (Lindane)	<0.00025	mg/L	0.00025	1	03/16/21 02:12	03/16/21 15:17	58-89-9	
Chlordane (Technical)	<0.0050	mg/L	0.0050	1	03/16/21 02:12	03/16/21 15:17	57-74-9	
Endrin	<0.00050	mg/L	0.00050	1	03/16/21 02:12	03/16/21 15:17	72-20-8	
Heptachlor	<0.00025	mg/L	0.00025	1	03/16/21 02:12	03/16/21 15:17	76-44-8	
Heptachlor epoxide	<0.00025	mg/L	0.00025	1	03/16/21 02:12	03/16/21 15:17	1024-57-3	
Methoxychlor	<0.0025	mg/L	0.0025	1	03/16/21 02:12	03/16/21 15:17	72-43-5	
Toxaphene	<0.0050	mg/L	0.0050	1	03/16/21 02:12	03/16/21 15:17	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	93	%.	10-113	1	03/16/21 02:12	03/16/21 15:17	2051-24-3	

8082 PCB Solids

Analytical Method: EPA 8082 Preparation Method: EPA 3546

Pace Analytical Services - Indianapolis

PCB-1016 (Aroclor 1016)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	12674-11-2	
PCB-1221 (Aroclor 1221)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	11104-28-2	
PCB-1232 (Aroclor 1232)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	11141-16-5	
PCB-1242 (Aroclor 1242)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	53469-21-9	
PCB-1248 (Aroclor 1248)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	12672-29-6	
PCB-1254 (Aroclor 1254)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	11097-69-1	
PCB-1260 (Aroclor 1260)	<9260	ug/kg	9260	1	03/17/21 09:47	03/18/21 12:34	11096-82-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8082 PCB Solids								
Analytical Method: EPA 8082 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Surrogates								
Tetrachloro-m-xylene (S)	72	%.	25-123	1	03/17/21 09:47	03/18/21 12:34	877-09-8	
8151A CI Acid Herbicides TCLP								
Analytical Method: EPA 8151A Preparation Method: EPA 8151								
Leachate Method/Date: EPA 1311; 03/10/21 14:00 Initial pH: 8.47; Final pH: 5.16								
Pace Analytical Services - Indianapolis								
2,4-D	<0.0050	mg/L	0.0050	1	03/17/21 11:42	03/17/21 18:03	94-75-7	
2,4,5-TP (Silvex)	<0.0050	mg/L	0.0050	1	03/17/21 11:42	03/17/21 18:03	93-72-1	
Surrogates								
2,4-DCAA (S)	75	%.	30-140	1	03/17/21 11:42	03/17/21 18:03	19719-28-9	
6010 MET ICP								
Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Pace Analytical Services - Indianapolis								
Aluminum	5100	mg/kg	1420	1	03/14/21 13:54	03/15/21 08:53	7429-90-5	
Calcium	22400	mg/kg	1420	1	03/14/21 13:54	03/15/21 08:53	7440-70-2	
Magnesium	4060	mg/kg	1420	1	03/14/21 13:54	03/15/21 08:53	7439-95-4	
Phosphorus	38500	mg/kg	1420	1	03/14/21 13:54	03/15/21 08:53	7723-14-0	N2
Potassium	2070	mg/kg	1420	1	03/14/21 13:54	03/15/21 08:53	7440-09-7	
Sodium	3450	mg/kg	1420	1	03/14/21 13:54	03/15/21 08:53	7440-23-5	
6010 MET ICP, TCLP								
Analytical Method: EPA 6010 Preparation Method: EPA 3010								
Leachate Method/Date: EPA 1311; 03/10/21 14:00 Initial pH: 8.47; Final pH: 5.16								
Pace Analytical Services - Indianapolis								
Arsenic	<0.10	mg/L	0.10	1	03/12/21 06:15	03/14/21 13:35	7440-38-2	
Barium	<5.0	mg/L	5.0	1	03/12/21 06:15	03/14/21 13:35	7440-39-3	
Cadmium	<0.050	mg/L	0.050	1	03/12/21 06:15	03/14/21 13:35	7440-43-9	
Chromium	<0.10	mg/L	0.10	1	03/12/21 06:15	03/14/21 13:35	7440-47-3	
Lead	<0.10	mg/L	0.10	1	03/12/21 06:15	03/14/21 13:35	7439-92-1	
Selenium	<0.10	mg/L	0.10	1	03/12/21 06:15	03/14/21 13:35	7782-49-2	
Silver	<0.10	mg/L	0.10	1	03/12/21 06:15	03/14/21 13:35	7440-22-4	
Zinc	3.3	mg/L	0.20	1	03/12/21 06:15	03/14/21 13:35	7440-66-6	
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Pace Analytical Services - Indianapolis								
Arsenic	16.9	mg/kg	3.0	1	03/10/21 08:37	03/11/21 07:07	7440-38-2	
Cadmium	1.6	mg/kg	1.5	1	03/10/21 08:37	03/11/21 07:07	7440-43-9	
Chromium	89.5	mg/kg	6.0	1	03/10/21 08:37	03/11/21 07:07	7440-47-3	
Copper	855	mg/kg	7.5	5	03/10/21 08:37	03/12/21 00:29	7440-50-8	
Lead	18.2	mg/kg	3.0	1	03/10/21 08:37	03/11/21 07:07	7439-92-1	
Molybdenum	20.6	mg/kg	3.0	1	03/10/21 08:37	03/11/21 07:07	7439-98-7	N2
Nickel	20.6	mg/kg	1.5	1	03/10/21 08:37	03/11/21 07:07	7440-02-0	
Selenium	4.4	mg/kg	3.0	1	03/10/21 08:37	03/11/21 07:07	7782-49-2	
Silver	2.1	mg/kg	1.5	1	03/10/21 08:37	03/11/21 07:07	7440-22-4	
Zinc	951	mg/kg	75.0	5	03/10/21 08:37	03/12/21 00:29	7440-66-6	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP								
Analytical Method: EPA 7470 Preparation Method: EPA 7470								
Leachate Method/Date: EPA 1311; 03/10/21 14:00 Initial pH: 8.47; Final pH: 5.16								
Pace Analytical Services - Indianapolis								
Mercury	<0.0020	mg/L	0.0020	1	03/12/21 09:46	03/15/21 11:34	7439-97-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Pace Analytical Services - Indianapolis								
Mercury	<1.3	mg/kg	1.3	1	03/11/21 01:32	03/11/21 09:40	7439-97-6	
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Acenaphthene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	83-32-9	L1
Acenaphthylene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	208-96-8	L1
Anthracene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	120-12-7	L1
Benzo(a)anthracene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	56-55-3	L1
Benzo(a)pyrene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	50-32-8	L1
Benzo(b)fluoranthene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	205-99-2	L1
Benzo(g,h,i)perylene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	191-24-2	
Benzo(k)fluoranthene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	207-08-9	
Benzyl alcohol	<106000	ug/kg	106000	1	03/17/21 12:00	03/17/21 17:03	100-51-6	
4-Bromophenylphenyl ether	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	101-55-3	
Butylbenzylphthalate	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	85-68-7	
4-Chloro-3-methylphenol	<106000	ug/kg	106000	1	03/17/21 12:00	03/17/21 17:03	132-64-9	
4-Chloroaniline	<106000	ug/kg	106000	1	03/17/21 12:00	03/17/21 17:03	106-47-8	
bis(2-Chloroethoxy)methane	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	111-91-1	
bis(2-Chloroethyl) ether	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	111-44-4	
bis(2chloro1methylethyl) ether	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	108-60-1	
2-Chloronaphthalene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	91-58-7	
2-Chlorophenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	95-57-8	L1
4-Chlorophenylphenyl ether	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	7005-72-3	
Chrysene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	218-01-9	L1
Dibenz(a,h)anthracene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	53-70-3	
Dibenzofuran	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	132-64-9	
3,3'-Dichlorobenzidine	<106000	ug/kg	106000	1	03/17/21 12:00	03/17/21 17:03	91-94-1	
2,4-Dichlorophenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	120-83-2	
Diethylphthalate	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	84-66-2	
2,4-Dimethylphenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	105-67-9	
Dimethylphthalate	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	131-11-3	
Di-n-butylphthalate	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	84-74-2	
4,6-Dinitro-2-methylphenol	<106000	ug/kg	106000	1	03/17/21 12:00	03/17/21 17:03	534-52-1	
2,4-Dinitrophenol	<256000	ug/kg	256000	1	03/17/21 12:00	03/17/21 17:03	51-28-5	
2,4-Dinitrotoluene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	121-14-2	
2,6-Dinitrotoluene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	606-20-2	
Di-n-octylphthalate	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	117-84-0	
bis(2-Ethylhexyl)phthalate	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	117-81-7	
Fluoranthene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	206-44-0	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Fluorene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	86-73-7	L1
Hexachloro-1,3-butadiene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	87-68-3	
Hexachlorobenzene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	118-74-1	
Hexachlorocyclopentadiene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	77-47-4	
Hexachloroethane	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	67-72-1	
Indeno(1,2,3-cd)pyrene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	193-39-5	
Isophorone	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	78-59-1	
1-Methylnaphthalene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	90-12-0	
2-Methylnaphthalene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	91-57-6	
2-Methylphenol(o-Cresol)	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	95-48-7	
3&4-Methylphenol(m&p Cresol)	<106000	ug/kg	106000	1	03/17/21 12:00	03/17/21 17:03		
Naphthalene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	91-20-3	L1
2-Nitroaniline	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	88-74-4	
3-Nitroaniline	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	99-09-2	
4-Nitroaniline	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	100-01-6	
Nitrobenzene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	98-95-3	
2-Nitrophenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	88-75-5	
4-Nitrophenol	<256000	ug/kg	256000	1	03/17/21 12:00	03/17/21 17:03	100-02-7	L1
N-Nitroso-di-n-propylamine	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	621-64-7	
N-Nitrosodiphenylamine	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	86-30-6	
Pentachlorophenol	<256000	ug/kg	256000	1	03/17/21 12:00	03/17/21 17:03	87-86-5	
Phenanthrene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	85-01-8	L1
Phenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	108-95-2	L1,P1
Pyrene	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	129-00-0	L1
2,4,5-Trichlorophenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	95-95-4	
2,4,6-Trichlorophenol	<52900	ug/kg	52900	1	03/17/21 12:00	03/17/21 17:03	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	51	%.	26-96	1	03/17/21 12:00	03/17/21 17:03	4165-60-0	
Phenol-d5 (S)	53	%.	32-93	1	03/17/21 12:00	03/17/21 17:03	4165-62-2	
2-Fluorophenol (S)	51	%.	24-95	1	03/17/21 12:00	03/17/21 17:03	367-12-4	
2,4,6-Tribromophenol (S)	70	%.	20-109	1	03/17/21 12:00	03/17/21 17:03	118-79-6	
2-Fluorobiphenyl (S)	52	%.	36-91	1	03/17/21 12:00	03/17/21 17:03	321-60-8	
p-Terphenyl-d14 (S)	93	%.	27-117	1	03/17/21 12:00	03/17/21 17:03	1718-51-0	

8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 03/10/21 14:00 Initial pH: 8.47; Final pH: 5.16

Pace Analytical Services - Indianapolis

1,4-Dichlorobenzene	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	106-46-7	
2,4-Dinitrotoluene	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	121-14-2	
Hexachloro-1,3-butadiene	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	87-68-3	
Hexachlorobenzene	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	118-74-1	L1
Hexachloroethane	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	67-72-1	
2-Methylphenol(o-Cresol)	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	95-48-7	
3&4-Methylphenol(m&p Cresol)	<0.20	mg/L	0.20	1	03/15/21 23:04	03/16/21 15:30		
Nitrobenzene	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	98-95-3	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
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8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 03/10/21 14:00 Initial pH: 8.47; Final pH: 5.16

Pace Analytical Services - Indianapolis

Pentachlorophenol	<0.50	mg/L	0.50	1	03/15/21 23:04	03/16/21 15:30	87-86-5	
Pyridine	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	110-86-1	
2,4,5-Trichlorophenol	<0.50	mg/L	0.50	1	03/15/21 23:04	03/16/21 15:30	95-95-4	
2,4,6-Trichlorophenol	<0.10	mg/L	0.10	1	03/15/21 23:04	03/16/21 15:30	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	71	%	49-100	1	03/15/21 23:04	03/16/21 15:30	4165-60-0	
2-Fluorobiphenyl (S)	75	%	42-95	1	03/15/21 23:04	03/16/21 15:30	321-60-8	
p-Terphenyl-d14 (S)	83	%	52-124	1	03/15/21 23:04	03/16/21 15:30	1718-51-0	
Phenol-d5 (S)	29	%	10-51	1	03/15/21 23:04	03/16/21 15:30	4165-62-2	
2-Fluorophenol (S)	45	%	23-70	1	03/15/21 23:04	03/16/21 15:30	367-12-4	
2,4,6-Tribromophenol (S)	91	%	54-106	1	03/15/21 23:04	03/16/21 15:30	118-79-6	

8260 MSV 5030 Low Level

Analytical Method: EPA 8260

Pace Analytical Services - Indianapolis

Acetone	<15800	mg/kg	15800	5000		03/16/21 19:45	67-64-1	
Acrolein	<15800	mg/kg	15800	5000		03/16/21 19:45	107-02-8	
Acrylonitrile	<15800	mg/kg	15800	5000		03/16/21 19:45	107-13-1	
Benzene	<788	mg/kg	788	5000		03/16/21 19:45	71-43-2	
Bromodichloromethane	<788	mg/kg	788	5000		03/16/21 19:45	75-27-4	
Bromoform	<788	mg/kg	788	5000		03/16/21 19:45	75-25-2	
Bromomethane	<788	mg/kg	788	5000		03/16/21 19:45	74-83-9	
2-Butanone (MEK)	<3940	mg/kg	3940	5000		03/16/21 19:45	78-93-3	
Carbon disulfide	<1580	mg/kg	1580	5000		03/16/21 19:45	75-15-0	
Carbon tetrachloride	<788	mg/kg	788	5000		03/16/21 19:45	56-23-5	
Chlorobenzene	<788	mg/kg	788	5000		03/16/21 19:45	108-90-7	
Chloroethane	<788	mg/kg	788	5000		03/16/21 19:45	75-00-3	
Chloroform	<788	mg/kg	788	5000		03/16/21 19:45	67-66-3	
Chloromethane	<788	mg/kg	788	5000		03/16/21 19:45	74-87-3	
1,2-Dibromo-3-chloropropane	<1580	mg/kg	1580	5000		03/16/21 19:45	96-12-8	
Dibromochloromethane	<788	mg/kg	788	5000		03/16/21 19:45	124-48-1	
1,2-Dibromoethane (EDB)	<788	mg/kg	788	5000		03/16/21 19:45	106-93-4	
Dibromomethane	<788	mg/kg	788	5000		03/16/21 19:45	74-95-3	
1,2-Dichlorobenzene	<788	mg/kg	788	5000		03/16/21 19:45	95-50-1	
1,3-Dichlorobenzene	<788	mg/kg	788	5000		03/16/21 19:45	541-73-1	
1,4-Dichlorobenzene	<788	mg/kg	788	5000		03/16/21 19:45	106-46-7	
trans-1,4-Dichloro-2-butene	<15800	mg/kg	15800	5000		03/16/21 19:45	110-57-6	
Dichlorodifluoromethane	<788	mg/kg	788	5000		03/16/21 19:45	75-71-8	
1,1-Dichloroethane	<788	mg/kg	788	5000		03/16/21 19:45	75-34-3	
1,2-Dichloroethane	<788	mg/kg	788	5000		03/16/21 19:45	107-06-2	
1,1-Dichloroethene	<788	mg/kg	788	5000		03/16/21 19:45	75-35-4	
cis-1,2-Dichloroethene	<788	mg/kg	788	5000		03/16/21 19:45	156-59-2	
trans-1,2-Dichloroethene	<788	mg/kg	788	5000		03/16/21 19:45	156-60-5	
1,2-Dichloropropane	<788	mg/kg	788	5000		03/16/21 19:45	78-87-5	
cis-1,3-Dichloropropene	<788	mg/kg	788	5000		03/16/21 19:45	10061-01-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level		Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis						
trans-1,3-Dichloropropene	<788	mg/kg	788	5000		03/16/21 19:45	10061-02-6	
Ethylbenzene	<788	mg/kg	788	5000		03/16/21 19:45	100-41-4	
Ethyl methacrylate	<15800	mg/kg	15800	5000		03/16/21 19:45	97-63-2	
2-Hexanone	<15800	mg/kg	15800	5000		03/16/21 19:45	591-78-6	
Iodomethane	<15800	mg/kg	15800	5000		03/16/21 19:45	74-88-4	
Methylene Chloride	<3150	mg/kg	3150	5000		03/16/21 19:45	75-09-2	
Methyl methacrylate	<1580	mg/kg	1580	5000		03/16/21 19:45	80-62-6	
4-Methyl-2-pentanone (MIBK)	<3940	mg/kg	3940	5000		03/16/21 19:45	108-10-1	
Methyl-tert-butyl ether	<788	mg/kg	788	5000		03/16/21 19:45	1634-04-4	
Styrene	<788	mg/kg	788	5000		03/16/21 19:45	100-42-5	
1,1,1,2-Tetrachloroethane	<788	mg/kg	788	5000		03/16/21 19:45	630-20-6	
1,1,2,2-Tetrachloroethane	<788	mg/kg	788	5000		03/16/21 19:45	79-34-5	
Tetrachloroethene	<788	mg/kg	788	5000		03/16/21 19:45	127-18-4	
Toluene	<788	mg/kg	788	5000		03/16/21 19:45	108-88-3	
1,1,1-Trichloroethane	<788	mg/kg	788	5000		03/16/21 19:45	71-55-6	
1,1,2-Trichloroethane	<788	mg/kg	788	5000		03/16/21 19:45	79-00-5	
Trichloroethene	<788	mg/kg	788	5000		03/16/21 19:45	79-01-6	
Trichlorofluoromethane	<788	mg/kg	788	5000		03/16/21 19:45	75-69-4	
1,2,3-Trichloropropane	<788	mg/kg	788	5000		03/16/21 19:45	96-18-4	
Vinyl acetate	<15800	mg/kg	15800	5000		03/16/21 19:45	108-05-4	
Vinyl chloride	<788	mg/kg	788	5000		03/16/21 19:45	75-01-4	
Xylene (Total)	<1580	mg/kg	1580	5000		03/16/21 19:45	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	103	%.	73-133	5000		03/16/21 19:45	1868-53-7	1d
Toluene-d8 (S)	101	%.	73-130	5000		03/16/21 19:45	2037-26-5	
4-Bromofluorobenzene (S)	103	%.	55-129	5000		03/16/21 19:45	460-00-4	

8260 MSV TCLP

Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 03/12/21 16:26

Pace Analytical Services - Indianapolis

Benzene	<0.050	mg/L	0.050	1		03/17/21 09:48	71-43-2	
2-Butanone (MEK)	<1.0	mg/L	1.0	1		03/17/21 09:48	78-93-3	
Carbon tetrachloride	<0.050	mg/L	0.050	1		03/17/21 09:48	56-23-5	
Chlorobenzene	<0.050	mg/L	0.050	1		03/17/21 09:48	108-90-7	
Chloroform	<0.050	mg/L	0.050	1		03/17/21 09:48	67-66-3	
1,2-Dichloroethane	<0.050	mg/L	0.050	1		03/17/21 09:48	107-06-2	
1,1-Dichloroethene	<0.050	mg/L	0.050	1		03/17/21 09:48	75-35-4	
Tetrachloroethene	<0.050	mg/L	0.050	1		03/17/21 09:48	127-18-4	M1
Trichloroethene	<0.050	mg/L	0.050	1		03/17/21 09:48	79-01-6	
Vinyl chloride	<0.020	mg/L	0.020	1		03/17/21 09:48	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	100	%.	85-116	1		03/17/21 09:48	460-00-4	
Dibromofluoromethane (S)	106	%.	75-120	1		03/17/21 09:48	1868-53-7	
Toluene-d8 (S)	99	%.	83-111	1		03/17/21 09:48	2037-26-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50281333

Sample: Digester #4 **Lab ID: 50281333001** Collected: 03/03/21 13:45 Received: 03/04/21 12:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
Percent Moisture								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Percent Moisture	96.8	%	0.10	1		03/09/21 12:02		N2
2540G Total Percent Solids								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Total Solids	3.2	%	0.10	1		03/09/21 12:02		N2
2540G Total Volatile Solids								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Total Volatile Solids	56.2	%	0.10	1		03/09/21 12:02		N2
9038 Sulfate Soil								
Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis								
Sulfate	<3150	mg/kg	3150	1	03/11/21 09:00	03/11/21 13:56	14808-79-8	N2
9045 pH Soil								
Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis								
pH at 25 Degrees C	7.8	Std. Units	0.10	1		03/08/21 13:55		H3
351.2 Total Kjeldahl Nitrogen								
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Green Bay								
Nitrogen, Kjeldahl, Total	63300	mg/kg	5940	2	03/15/21 14:40	03/15/21 19:51	7727-37-9	
353.2 Nitrogen, NO2/NO3								
Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis								
Nitrogen, NO2 plus NO3	<157	mg/kg	157	1	03/09/21 09:43	03/09/21 12:10		B0,N2
Nitrogen, Nitrate	<157	mg/kg	157	1	03/09/21 09:43	03/09/21 12:10	14797-55-8	B0,N2
4500 Ammonia Soil, Distilled								
Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis								
Nitrogen, Ammonia	27900	mg/kg	1560	10	03/11/21 11:32	03/11/21 14:07	7664-41-7	N2
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis								
Chloride	4900	mg/kg	3150	1	03/12/21 06:41	03/12/21 08:14	16887-00-6	N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609918

QC Batch Method: EPA 7470

Analysis Method: EPA 7470

Analysis Description: 7470 Mercury TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2810987

Matrix: Water

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	03/15/21 11:00	

LABORATORY CONTROL SAMPLE: 2810988

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0049	98	80-120	

MATRIX SPIKE SAMPLE: 2810989

Parameter	Units	50281294001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	94	75-125	

MATRIX SPIKE SAMPLE: 2810990

Parameter	Units	50281296001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	96	75-125	

MATRIX SPIKE SAMPLE: 2810991

Parameter	Units	50281303001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	98	75-125	

MATRIX SPIKE SAMPLE: 2810992

Parameter	Units	50281363001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.013	86	75-125	

MATRIX SPIKE SAMPLE: 2810993

Parameter	Units	50281390001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	96	75-125	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

MATRIX SPIKE SAMPLE:		2810994					
Parameter	Units	50281333001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.014	94	75-125	

MATRIX SPIKE SAMPLE:		2810995					
Parameter	Units	50282014001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	97	75-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE:		2810996		2810997								
Parameter	Units	50281845002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	ND	0.015	0.015	0.014	0.014	92	95	75-125	3	20	

MATRIX SPIKE SAMPLE:		2810998					
Parameter	Units	50281847002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	98	75-125	

MATRIX SPIKE SAMPLE:		2810999					
Parameter	Units	50281848002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.013	85	75-125	

MATRIX SPIKE SAMPLE:		2811000					
Parameter	Units	50281849002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	95	75-125	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch: 609472	Analysis Method: EPA 7471
QC Batch Method: EPA 7471	Analysis Description: 7471 Mercury
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2808783 Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.21	0.21	03/11/21 09:30	

LABORATORY CONTROL SAMPLE: 2808784

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.52	0.51	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2808785 2808786

Parameter	Units	50281361002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.75	0.76	0.86	0.86	100	97	75-125	1	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609624

Analysis Method: EPA 6010

QC Batch Method: EPA 3050

Analysis Description: 6010 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2809627

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<46.7	46.7	03/15/21 08:42	
Calcium	mg/kg	<46.7	46.7	03/15/21 08:42	
Magnesium	mg/kg	<46.7	46.7	03/15/21 08:42	
Phosphorus	mg/kg	ND	46.7	03/15/21 08:42	N2
Potassium	mg/kg	<46.7	46.7	03/15/21 08:42	
Sodium	mg/kg	<46.7	46.7	03/15/21 08:42	

LABORATORY CONTROL SAMPLE: 2809628

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	483	477	99	80-120	
Calcium	mg/kg	483	512	106	80-120	
Magnesium	mg/kg	483	476	99	80-120	
Phosphorus	mg/kg	48.3	49.2	102	80-120	N2
Potassium	mg/kg	483	488	101	80-120	
Sodium	mg/kg	483	488	101	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2809629 2809630

Parameter	Units	50281766011 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	14200	608	603	18100	18900	633	779	75-125	5	20	E,P6
Calcium	mg/kg	2930	608	603	3550	3990	103	177	75-125	12	20	P6
Magnesium	mg/kg	2550	608	603	3340	3510	131	160	75-125	5	20	P6
Phosphorus	mg/kg	391	60.8	60.3	449	429	96	63	75-125	5	20	N2,P6
Potassium	mg/kg	1150	608	603	2120	2160	161	169	75-125	2	20	M3
Sodium	mg/kg	ND	608	603	581	592	89	91	75-125	2	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609722

Analysis Method: EPA 6010

QC Batch Method: EPA 3010

Analysis Description: 6010 MET TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2810045

Matrix: Water

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	03/14/21 13:04	
Barium	mg/L	<0.50	0.50	03/14/21 13:04	
Cadmium	mg/L	<0.0050	0.0050	03/14/21 13:04	
Chromium	mg/L	<0.010	0.010	03/14/21 13:04	
Lead	mg/L	<0.010	0.010	03/14/21 13:04	
Selenium	mg/L	<0.010	0.010	03/14/21 13:04	
Silver	mg/L	<0.010	0.010	03/14/21 13:04	
Zinc	mg/L	<0.020	0.020	03/14/21 13:04	

LABORATORY CONTROL SAMPLE: 2810046

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	1.0	102	80-120	
Barium	mg/L	1	1.0	100	80-120	
Cadmium	mg/L	1	0.97	97	80-120	
Chromium	mg/L	1	1.0	102	80-120	
Lead	mg/L	1	0.96	96	80-120	
Selenium	mg/L	1	0.99	99	80-120	
Silver	mg/L	0.5	0.50	101	80-120	
Zinc	mg/L	1	0.99	99	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2810047 2810048

Parameter	Units	50281293001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	6.4	7.5	64	75	50-150	15	20	
Barium	mg/L	ND	10	10	10.2	10.4	102	104	50-150	2	20	
Cadmium	mg/L	ND	10	10	10.3	10.5	103	105	50-150	2	20	
Chromium	mg/L	ND	10	10	10.6	10.8	106	108	50-150	2	20	
Lead	mg/L	ND	10	10	9.5	9.7	95	97	50-150	2	20	
Selenium	mg/L	ND	10	10	10.6	10.8	105	108	50-150	2	20	
Silver	mg/L	ND	5	5	4.9	4.9	98	98	50-150	1	20	
Zinc	mg/L	ND	10	10	10.6	10.8	106	108	50-150	2	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

MATRIX SPIKE SAMPLE:		2810049					
Parameter	Units	50281294001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.2	102	50-150	
Barium	mg/L	ND	10	10.2	101	50-150	
Cadmium	mg/L	ND	10	9.8	98	50-150	
Chromium	mg/L	ND	10	10.3	103	50-150	
Lead	mg/L	ND	10	9.6	96	50-150	
Selenium	mg/L	ND	10	10	100	50-150	
Silver	mg/L	ND	5	5.1	101	50-150	
Zinc	mg/L	ND	10	10.0	100	50-150	

MATRIX SPIKE SAMPLE:		2810050					
Parameter	Units	50281296001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.6	106	50-150	
Barium	mg/L	ND	10	10.1	100	50-150	
Cadmium	mg/L	ND	10	9.9	99	50-150	
Chromium	mg/L	220	10	223	36	50-150	P6
Lead	mg/L	ND	10	9.1	90	50-150	
Selenium	mg/L	ND	10	10.3	102	50-150	
Silver	mg/L	ND	5	5.1	102	50-150	
Zinc	mg/L	14.1	10	23.8	97	50-150	

MATRIX SPIKE SAMPLE:		2810051					
Parameter	Units	50281363001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.5	105	50-150	
Barium	mg/L	ND	10	11.6	102	50-150	
Cadmium	mg/L	ND	10	10	99	50-150	
Chromium	mg/L	16.2	10	26.4	102	50-150	
Lead	mg/L	4.3	10	13.6	93	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	
Silver	mg/L	ND	5	5.1	102	50-150	
Zinc	mg/L	909000 ug/L	10	931	224	50-150	E,P6

MATRIX SPIKE SAMPLE:		2810052					
Parameter	Units	50281333001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.6	105	50-150	
Barium	mg/L	<5.0	10	10.3	101	50-150	
Cadmium	mg/L	<0.050	10	9.8	98	50-150	
Chromium	mg/L	<0.10	10	10.2	102	50-150	
Lead	mg/L	<0.10	10	9.4	94	50-150	
Selenium	mg/L	<0.10	10	10.2	102	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

MATRIX SPIKE SAMPLE:		2810052					
		50281333001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Silver	mg/L	<0.10	5	5.1	101	50-150	
Zinc	mg/L	3.3	10	13.2	100	50-150	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609244

Analysis Method: EPA 6020

QC Batch Method: EPA 3050B

Analysis Description: 6020 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2807720

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.097	0.097	03/11/21 06:49	
Cadmium	mg/kg	<0.048	0.048	03/11/21 06:49	
Chromium	mg/kg	<0.19	0.19	03/11/21 06:49	
Copper	mg/kg	<0.048	0.048	03/11/21 06:49	
Lead	mg/kg	<0.097	0.097	03/11/21 06:49	
Molybdenum	mg/kg	<0.097	0.097	03/11/21 06:49	N2
Nickel	mg/kg	<0.048	0.048	03/11/21 06:49	
Selenium	mg/kg	<0.097	0.097	03/11/21 06:49	
Silver	mg/kg	<0.048	0.048	03/11/21 06:49	
Zinc	mg/kg	<0.48	0.48	03/11/21 06:49	

LABORATORY CONTROL SAMPLE: 2807721

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	3.9	3.3	85	80-120	
Cadmium	mg/kg	3.9	3.5	89	80-120	
Chromium	mg/kg	3.9	3.7	96	80-120	
Copper	mg/kg	3.9	3.7	94	80-120	
Lead	mg/kg	3.9	3.9	101	80-120	
Molybdenum	mg/kg	3.9	3.8	98	80-120	N2
Nickel	mg/kg	3.9	3.7	95	80-120	
Selenium	mg/kg	3.9	3.3	86	80-120	
Silver	mg/kg	3.9	3.8	98	80-120	
Zinc	mg/kg	3.9	3.3	85	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2807722 2807723

Parameter	Units	50281333001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	16.9	123	120	127	119	89	84	75-125	7	20	
Cadmium	mg/kg	1.6	123	120	115	113	91	92	75-125	1	20	
Chromium	mg/kg	89.5	123	120	232	218	115	106	75-125	6	20	
Copper	mg/kg	855	123	120	1110	1040	209	149	75-125	7	20	P6
Lead	mg/kg	18.2	123	120	131	127	91	90	75-125	3	20	
Molybdenum	mg/kg	20.6	123	120	149	140	104	99	75-125	6	20	N2
Nickel	mg/kg	20.6	123	120	144	138	100	98	75-125	4	20	
Selenium	mg/kg	4.4	123	120	110	106	85	84	75-125	4	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2807722 2807723												
Parameter	Units	50281333001 Result	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max	
			Spike Conc.	Spike Conc.							RPD	Qual
Silver	mg/kg	2.1	123	120	126	123	100	100	75-125	3	20	
Zinc	mg/kg	951	123	120	1210	1170	211	182	75-125	4	20	P6

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 610552

Analysis Method: EPA 8260

QC Batch Method: EPA 8260

Analysis Description: 8260 MSV 5030 Low

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2814167

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	03/16/21 14:01	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
2-Butanone (MEK)	mg/kg	<0.025	0.025	03/16/21 14:01	
2-Hexanone	mg/kg	<0.10	0.10	03/16/21 14:01	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	03/16/21 14:01	
Acetone	mg/kg	<0.10	0.10	03/16/21 14:01	
Acrolein	mg/kg	<0.10	0.10	03/16/21 14:01	
Acrylonitrile	mg/kg	<0.10	0.10	03/16/21 14:01	
Benzene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Bromodichloromethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Bromoform	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Bromomethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Carbon disulfide	mg/kg	<0.010	0.010	03/16/21 14:01	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Chlorobenzene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Chloroethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Chloroform	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Chloromethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Dibromochloromethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Dibromomethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Ethyl methacrylate	mg/kg	<0.10	0.10	03/16/21 14:01	
Ethylbenzene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Iodomethane	mg/kg	<0.10	0.10	03/16/21 14:01	
Methyl methacrylate	mg/kg	<0.010	0.010	03/16/21 14:01	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	03/16/21 14:01	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

METHOD BLANK: 2814167

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	03/16/21 14:01	
Styrene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Tetrachloroethene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Toluene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	03/16/21 14:01	
Trichloroethene	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Vinyl acetate	mg/kg	<0.10	0.10	03/16/21 14:01	
Vinyl chloride	mg/kg	<0.0050	0.0050	03/16/21 14:01	
Xylene (Total)	mg/kg	<0.010	0.010	03/16/21 14:01	
4-Bromofluorobenzene (S)	%	103	55-129	03/16/21 14:01	
Dibromofluoromethane (S)	%	104	73-133	03/16/21 14:01	
Toluene-d8 (S)	%	101	73-130	03/16/21 14:01	

LABORATORY CONTROL SAMPLE: 2814168

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1-Trichloroethane	mg/kg	0.05	0.050	100	72-132	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.048	96	63-127	
1,1-Dichloroethene	mg/kg	0.05	0.051	102	70-126	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.048	97	73-124	
1,2-Dichloroethane	mg/kg	0.05	0.052	103	67-129	
1,2-Dichloropropane	mg/kg	0.05	0.055	110	72-125	
Benzene	mg/kg	0.05	0.052	103	74-117	
Chlorobenzene	mg/kg	0.05	0.048	96	72-114	
Chloroform	mg/kg	0.05	0.049	99	69-120	
cis-1,2-Dichloroethene	mg/kg	0.05	0.049	98	73-117	
Ethylbenzene	mg/kg	0.05	0.050	100	68-118	
Methyl-tert-butyl ether	mg/kg	0.05	0.051	102	75-136	
Tetrachloroethene	mg/kg	0.05	0.048	96	59-119	
Toluene	mg/kg	0.05	0.049	98	68-115	
trans-1,2-Dichloroethene	mg/kg	0.05	0.050	100	71-125	
Trichloroethene	mg/kg	0.05	0.051	102	68-118	
Vinyl chloride	mg/kg	0.05	0.063	125	49-125	
Xylene (Total)	mg/kg	0.15	0.14	96	69-118	
4-Bromofluorobenzene (S)	%			102	55-129	
Dibromofluoromethane (S)	%			95	73-133	
Toluene-d8 (S)	%			102	73-130	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 610542

Analysis Method: EPA 5030/8260

QC Batch Method: EPA 5030/8260

Analysis Description: 8260 MSV TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2814058

Matrix: Water

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	03/17/21 01:11	
1,2-Dichloroethane	mg/L	<0.050	0.050	03/17/21 01:11	
2-Butanone (MEK)	mg/L	<1.0	1.0	03/17/21 01:11	
Benzene	mg/L	<0.050	0.050	03/17/21 01:11	
Carbon tetrachloride	mg/L	<0.050	0.050	03/17/21 01:11	
Chlorobenzene	mg/L	<0.050	0.050	03/17/21 01:11	
Chloroform	mg/L	<0.050	0.050	03/17/21 01:11	
Tetrachloroethene	mg/L	<0.050	0.050	03/17/21 01:11	
Trichloroethene	mg/L	<0.050	0.050	03/17/21 01:11	
Vinyl chloride	mg/L	<0.020	0.020	03/17/21 01:11	
4-Bromofluorobenzene (S)	%	99	85-116	03/17/21 01:11	
Dibromofluoromethane (S)	%	103	75-120	03/17/21 01:11	
Toluene-d8 (S)	%	99	83-111	03/17/21 01:11	

LABORATORY CONTROL SAMPLE: 2814059

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.48	96	79-128	
1,2-Dichloroethane	mg/L	0.5	0.52	105	66-127	
2-Butanone (MEK)	mg/L	2.5	2.5	98	61-138	
Benzene	mg/L	0.5	0.49	97	75-118	
Carbon tetrachloride	mg/L	0.5	0.49	98	73-125	
Chlorobenzene	mg/L	0.5	0.47	94	80-115	
Chloroform	mg/L	0.5	0.50	99	75-117	
Tetrachloroethene	mg/L	0.5	0.49	97	70-123	
Trichloroethene	mg/L	0.5	0.50	100	78-120	
Vinyl chloride	mg/L	0.5	0.53	106	55-122	
4-Bromofluorobenzene (S)	%			103	85-116	
Dibromofluoromethane (S)	%			100	75-120	
Toluene-d8 (S)	%			100	83-111	

MATRIX SPIKE SAMPLE: 2814060

Parameter	Units	50281946001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.50	101	60-140	
1,2-Dichloroethane	mg/L	ND	0.5	0.58	116	46-145	
2-Butanone (MEK)	mg/L	ND	2.5	2.7	107	42-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

MATRIX SPIKE SAMPLE:		2814060					
Parameter	Units	50281946001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	ND	0.5	0.52	103	49-135	
Carbon tetrachloride	mg/L	ND	0.5	0.52	103	55-134	
Chlorobenzene	mg/L	ND	0.5	0.48	95	42-135	
Chloroform	mg/L	ND	0.5	0.54	107	57-130	
Tetrachloroethene	mg/L	ND	0.5	0.48	96	32-140	
Trichloroethene	mg/L	ND	0.5	0.51	102	47-137	
Vinyl chloride	mg/L	ND	0.5	0.57	113	36-136	
4-Bromofluorobenzene (S)	%				104	85-116	
Dibromofluoromethane (S)	%				104	75-120	
Toluene-d8 (S)	%				101	83-111	

MATRIX SPIKE SAMPLE:		2814061					
Parameter	Units	50281333001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.5	0.53	107	60-140	
1,2-Dichloroethane	mg/L	<0.050	0.5	0.57	114	46-145	
2-Butanone (MEK)	mg/L	<1.0	2.5	2.5	99	42-150	
Benzene	mg/L	<0.050	0.5	0.54	108	49-135	
Carbon tetrachloride	mg/L	<0.050	0.5	0.55	110	55-134	
Chlorobenzene	mg/L	<0.050	0.5	0.49	98	42-135	
Chloroform	mg/L	<0.050	0.5	0.55	111	57-130	
Tetrachloroethene	mg/L	<0.050	0.5	0.76	152	32-140 M1	
Trichloroethene	mg/L	<0.050	0.5	0.55	110	47-137	
Vinyl chloride	mg/L	<0.020	0.5	0.59	119	36-136	
4-Bromofluorobenzene (S)	%				101	85-116	
Dibromofluoromethane (S)	%				100	75-120	
Toluene-d8 (S)	%				100	83-111	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609930

Analysis Method: EPA 8081

QC Batch Method: EPA 3546

Analysis Description: 8081 GCS Pesticides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2811070

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	03/15/21 12:02	
4,4'-DDE	ug/kg	<5.0	5.0	03/15/21 12:02	
4,4'-DDT	ug/kg	<5.0	5.0	03/15/21 12:02	
Aldrin	ug/kg	<2.5	2.5	03/15/21 12:02	
alpha-BHC	ug/kg	<2.5	2.5	03/15/21 12:02	
alpha-Chlordane	ug/kg	<2.5	2.5	03/15/21 12:02	
beta-BHC	ug/kg	<2.5	2.5	03/15/21 12:02	
Chlordane (Technical)	ug/kg	<49.5	49.5	03/15/21 12:02	
delta-BHC	ug/kg	<2.5	2.5	03/15/21 12:02	
Dieldrin	ug/kg	<5.0	5.0	03/15/21 12:02	
Endosulfan I	ug/kg	<2.5	2.5	03/15/21 12:02	
Endosulfan II	ug/kg	<5.0	5.0	03/15/21 12:02	
Endosulfan sulfate	ug/kg	<5.0	5.0	03/15/21 12:02	
Endrin	ug/kg	<5.0	5.0	03/15/21 12:02	
Endrin aldehyde	ug/kg	<5.0	5.0	03/15/21 12:02	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	03/15/21 12:02	
gamma-Chlordane	ug/kg	<2.5	2.5	03/15/21 12:02	
Heptachlor	ug/kg	<2.5	2.5	03/15/21 12:02	
Heptachlor epoxide	ug/kg	<2.5	2.5	03/15/21 12:02	
Toxaphene	ug/kg	<49.5	49.5	03/15/21 12:02	
Decachlorobiphenyl (S)	%	103	10-129	03/15/21 12:02	

LABORATORY CONTROL SAMPLE: 2811071

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	20	14.8	74	53-155	
4,4'-DDE	ug/kg	20	14.4	72	52-153	
4,4'-DDT	ug/kg	20	12.9	65	38-166	
Aldrin	ug/kg	10	6.8	68	49-148	
alpha-BHC	ug/kg	10	7.1	71	48-150	
alpha-Chlordane	ug/kg	10	8.7	87	52-147	
beta-BHC	ug/kg	10	8.3	83	50-150	
delta-BHC	ug/kg	10	5.6	56	35-127	
Dieldrin	ug/kg	20	14.4	72	52-144	
Endosulfan I	ug/kg	10	7.5	75	46-145	
Endosulfan II	ug/kg	20	15.0	75	52-146	
Endosulfan sulfate	ug/kg	20	16.6	83	50-137	
Endrin	ug/kg	20	14.3	71	48-149	
Endrin aldehyde	ug/kg	20	19.2	96	52-145	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

LABORATORY CONTROL SAMPLE: 2811071

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	10	6.9	69	47-154	
gamma-Chlordane	ug/kg	10	7.4	74	52-149	
Heptachlor	ug/kg	10	6.8	68	46-147	
Heptachlor epoxide	ug/kg	10	6.9	69	51-143	
Decachlorobiphenyl (S)	%			70	10-129	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2811072 2811073

Parameter	Units	50281361004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	26.7	26.8	18.9	20.8	71	77	18-157	9	20	
4,4'-DDE	ug/kg	ND	26.7	26.8	16.4	18.6	61	69	22-154	13	20	
4,4'-DDT	ug/kg	ND	26.7	26.8	18.7	20.9	70	78	10-183	11	20	
Aldrin	ug/kg	ND	13.3	13.5	8.4	9.7	63	73	21-154	15	20	
alpha-BHC	ug/kg	ND	13.3	13.5	9.4	10.2	70	76	22-156	9	20	
alpha-Chlordane	ug/kg	ND	13.3	13.5	11.0	11.7	83	87	12-163	6	20	
beta-BHC	ug/kg	ND	13.3	13.5	9.6	10.3	72	77	18-160	8	20	
delta-BHC	ug/kg	ND	13.3	13.5	7.8	8.4	59	62	10-145	7	20	
Dieldrin	ug/kg	ND	26.7	26.8	19.7	22.0	74	82	21-147	11	20	
Endosulfan I	ug/kg	ND	13.3	13.5	8.9	10.4	66	77	10-159	16	20	
Endosulfan II	ug/kg	ND	26.7	26.8	18.5	20.3	69	76	10-152	9	20	
Endosulfan sulfate	ug/kg	ND	26.7	26.8	18.4	20.3	69	76	10-145	10	20	
Endrin	ug/kg	ND	26.7	26.8	17.6	19.9	66	74	17-149	12	20	
Endrin aldehyde	ug/kg	ND	26.7	26.8	23.4	24.9	88	93	10-154	6	20	
gamma-BHC (Lindane)	ug/kg	ND	13.3	13.5	10.4	11.3	78	84	17-165	8	20	
gamma-Chlordane	ug/kg	ND	13.3	13.5	8.0	9.7	60	72	13-156	19	20	
Heptachlor	ug/kg	ND	13.3	13.5	9.6	10.3	72	77	12-164	8	20	
Heptachlor epoxide	ug/kg	ND	13.3	13.5	10.6	12.3	79	92	18-153	15	20	
Decachlorobiphenyl (S)	%						82	81	10-129			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 610318

Analysis Method: EPA 8081

QC Batch Method: EPA 3510

Analysis Description: 8081 GCS TCLP Pesticides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2813120

Matrix: Water

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.0010	0.0010	03/16/21 14:52	
Endrin	mg/L	<0.00010	0.00010	03/16/21 14:52	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	03/16/21 14:52	
Heptachlor	mg/L	<0.000050	0.000050	03/16/21 14:52	
Heptachlor epoxide	mg/L	<0.000050	0.000050	03/16/21 14:52	
Methoxychlor	mg/L	<0.00050	0.00050	03/16/21 14:52	
Toxaphene	mg/L	<0.0010	0.0010	03/16/21 14:52	
Decachlorobiphenyl (S)	%	92	10-113	03/16/21 14:52	

LABORATORY CONTROL SAMPLE: 2813121

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0018	89	40-160	
gamma-BHC (Lindane)	mg/L	0.001	0.00091	91	34-160	
Heptachlor	mg/L	0.001	0.00090	90	12-149	
Heptachlor epoxide	mg/L	0.001	0.00086	86	38-157	
Methoxychlor	mg/L	0.01	0.0092	92	38-163	
Decachlorobiphenyl (S)	%			62	10-113	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch: 610635	Analysis Method: EPA 8082
QC Batch Method: EPA 3546	Analysis Description: 8082 PCB Solids
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2814465 Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<99.3	99.3	03/17/21 11:38	
PCB-1221 (Aroclor 1221)	ug/kg	<99.3	99.3	03/17/21 11:38	
PCB-1232 (Aroclor 1232)	ug/kg	<99.3	99.3	03/17/21 11:38	
PCB-1242 (Aroclor 1242)	ug/kg	<99.3	99.3	03/17/21 11:38	
PCB-1248 (Aroclor 1248)	ug/kg	<99.3	99.3	03/17/21 11:38	
PCB-1254 (Aroclor 1254)	ug/kg	<99.3	99.3	03/17/21 11:38	
PCB-1260 (Aroclor 1260)	ug/kg	<99.3	99.3	03/17/21 11:38	
Tetrachloro-m-xylene (S)	%.	100	25-123	03/17/21 11:38	

LABORATORY CONTROL SAMPLE: 2814466

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	333	342	103	54-128	
PCB-1260 (Aroclor 1260)	ug/kg	333	267	80	36-118	
Tetrachloro-m-xylene (S)	%.			95	25-123	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch: 610667	Analysis Method: EPA 8151A
QC Batch Method: EPA 8151	Analysis Description: 8151 GCS TCLP Herbicides
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2814588 Matrix: Water

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	03/17/21 17:25	
2,4-D	mg/L	<0.0010	0.0010	03/17/21 17:25	
2,4-DCAA (S)	%.	89	30-140	03/17/21 17:25	

LABORATORY CONTROL SAMPLE: 2814589

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0036	71	41-141	
2,4-D	mg/L	0.005	0.0055	110	41-125	
2,4-DCAA (S)	%.			66	30-140	

MATRIX SPIKE SAMPLE: 2814590

Parameter	Units	50281946001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	ND	0.025	0.021	83	27-137	
2,4-D	mg/L	ND	0.025	0.026	103	22-128	
2,4-DCAA (S)	%.				65	30-140	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 610663

Analysis Method: EPA 8270

QC Batch Method: EPA 3546

Analysis Description: 8270 Solid MSSV Microwave Short Spike

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2814566

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<330	330	03/17/21 16:31	
2,4,5-Trichlorophenol	ug/kg	<330	330	03/17/21 16:31	
2,4,6-Trichlorophenol	ug/kg	<330	330	03/17/21 16:31	
2,4-Dichlorophenol	ug/kg	<330	330	03/17/21 16:31	
2,4-Dimethylphenol	ug/kg	<330	330	03/17/21 16:31	
2,4-Dinitrophenol	ug/kg	<1600	1600	03/17/21 16:31	
2,4-Dinitrotoluene	ug/kg	<330	330	03/17/21 16:31	
2,6-Dinitrotoluene	ug/kg	<330	330	03/17/21 16:31	
2-Chloronaphthalene	ug/kg	<330	330	03/17/21 16:31	
2-Chlorophenol	ug/kg	<330	330	03/17/21 16:31	
2-Methylnaphthalene	ug/kg	<330	330	03/17/21 16:31	
2-Methylphenol(o-Cresol)	ug/kg	<330	330	03/17/21 16:31	
2-Nitroaniline	ug/kg	<330	330	03/17/21 16:31	
2-Nitrophenol	ug/kg	<330	330	03/17/21 16:31	
3&4-Methylphenol(m&p Cresol)	ug/kg	<660	660	03/17/21 16:31	
3,3'-Dichlorobenzidine	ug/kg	<660	660	03/17/21 16:31	
3-Nitroaniline	ug/kg	<330	330	03/17/21 16:31	
4,6-Dinitro-2-methylphenol	ug/kg	<660	660	03/17/21 16:31	
4-Bromophenylphenyl ether	ug/kg	<330	330	03/17/21 16:31	
4-Chloro-3-methylphenol	ug/kg	<660	660	03/17/21 16:31	
4-Chloroaniline	ug/kg	<660	660	03/17/21 16:31	
4-Chlorophenylphenyl ether	ug/kg	<330	330	03/17/21 16:31	
4-Nitroaniline	ug/kg	<330	330	03/17/21 16:31	
4-Nitrophenol	ug/kg	<1600	1600	03/17/21 16:31	
Acenaphthene	ug/kg	<330	330	03/17/21 16:31	
Acenaphthylene	ug/kg	<330	330	03/17/21 16:31	
Anthracene	ug/kg	<330	330	03/17/21 16:31	
Benzo(a)anthracene	ug/kg	<330	330	03/17/21 16:31	
Benzo(a)pyrene	ug/kg	<330	330	03/17/21 16:31	
Benzo(b)fluoranthene	ug/kg	<330	330	03/17/21 16:31	
Benzo(g,h,i)perylene	ug/kg	<330	330	03/17/21 16:31	
Benzo(k)fluoranthene	ug/kg	<330	330	03/17/21 16:31	
Benzyl alcohol	ug/kg	<660	660	03/17/21 16:31	
bis(2-Chloroethoxy)methane	ug/kg	<330	330	03/17/21 16:31	
bis(2-Chloroethyl) ether	ug/kg	<330	330	03/17/21 16:31	
bis(2-Ethylhexyl)phthalate	ug/kg	<330	330	03/17/21 16:31	
bis(2chloro1methylethyl) ether	ug/kg	<330	330	03/17/21 16:31	
Butylbenzylphthalate	ug/kg	<330	330	03/17/21 16:31	
Chrysene	ug/kg	<330	330	03/17/21 16:31	
Di-n-butylphthalate	ug/kg	<330	330	03/17/21 16:31	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

METHOD BLANK: 2814566

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<330	330	03/17/21 16:31	
Dibenz(a,h)anthracene	ug/kg	<330	330	03/17/21 16:31	
Dibenzofuran	ug/kg	<330	330	03/17/21 16:31	
Diethylphthalate	ug/kg	<330	330	03/17/21 16:31	
Dimethylphthalate	ug/kg	<330	330	03/17/21 16:31	
Fluoranthene	ug/kg	<330	330	03/17/21 16:31	
Fluorene	ug/kg	<330	330	03/17/21 16:31	
Hexachloro-1,3-butadiene	ug/kg	<330	330	03/17/21 16:31	
Hexachlorobenzene	ug/kg	<330	330	03/17/21 16:31	
Hexachlorocyclopentadiene	ug/kg	<330	330	03/17/21 16:31	
Hexachloroethane	ug/kg	<330	330	03/17/21 16:31	
Indeno(1,2,3-cd)pyrene	ug/kg	<330	330	03/17/21 16:31	
Isophorone	ug/kg	<330	330	03/17/21 16:31	
N-Nitroso-di-n-propylamine	ug/kg	<330	330	03/17/21 16:31	
N-Nitrosodiphenylamine	ug/kg	<330	330	03/17/21 16:31	
Naphthalene	ug/kg	<330	330	03/17/21 16:31	
Nitrobenzene	ug/kg	<330	330	03/17/21 16:31	
Pentachlorophenol	ug/kg	<1600	1600	03/17/21 16:31	
Phenanthrene	ug/kg	<330	330	03/17/21 16:31	
Phenol	ug/kg	<330	330	03/17/21 16:31	
Pyrene	ug/kg	<330	330	03/17/21 16:31	
2,4,6-Tribromophenol (S)	%	66	20-109	03/17/21 16:31	
2-Fluorobiphenyl (S)	%	55	36-91	03/17/21 16:31	
2-Fluorophenol (S)	%	60	24-95	03/17/21 16:31	
Nitrobenzene-d5 (S)	%	52	26-96	03/17/21 16:31	
p-Terphenyl-d14 (S)	%	85	27-117	03/17/21 16:31	
Phenol-d5 (S)	%	58	32-93	03/17/21 16:31	

LABORATORY CONTROL SAMPLE: 2814567

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	664	710	107	41-116	
2-Chlorophenol	ug/kg	664	654	98	48-97	L1
2-Methylnaphthalene	ug/kg	664	628	94	51-98	
4-Chloro-3-methylphenol	ug/kg	664	705	106	51-108	
4-Nitrophenol	ug/kg	664	<1590	119	42-118	L1
Acenaphthene	ug/kg	664	664	100	58-94	L1
Acenaphthylene	ug/kg	664	708	107	56-94	L1
Anthracene	ug/kg	664	684	103	59-98	L1
Benzo(a)anthracene	ug/kg	664	719	108	60-103	L1
Benzo(a)pyrene	ug/kg	664	686	103	57-96	L1
Benzo(b)fluoranthene	ug/kg	664	763	115	57-104	L1
Benzo(g,h,i)perylene	ug/kg	664	680	102	56-103	
Benzo(k)fluoranthene	ug/kg	664	671	101	55-104	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

LABORATORY CONTROL SAMPLE: 2814567

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	664	705	106	59-105	L1
Dibenz(a,h)anthracene	ug/kg	664	642	97	54-103	
Fluoranthene	ug/kg	664	706	106	57-107	
Fluorene	ug/kg	664	700	105	58-101	L1
Indeno(1,2,3-cd)pyrene	ug/kg	664	672	101	54-104	
N-Nitroso-di-n-propylamine	ug/kg	664	621	93	43-94	
Naphthalene	ug/kg	664	653	98	51-90	L1
Pentachlorophenol	ug/kg	664	<1590	86	24-118	
Phenanthrene	ug/kg	664	717	108	60-96	L1
Phenol	ug/kg	664	705	106	44-101	L1
Pyrene	ug/kg	664	756	114	59-108	L1
2,4,6-Tribromophenol (S)	%			49	20-109	
2-Fluorobiphenyl (S)	%			39	36-91	
2-Fluorophenol (S)	%			45	24-95	
Nitrobenzene-d5 (S)	%			38	26-96	
p-Terphenyl-d14 (S)	%			58	27-117	
Phenol-d5 (S)	%			42	32-93	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2814568 2814569

Parameter	Units	5263625005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	146000	133000	173000	167000	119	126	10-127	4	20	
2-Chlorophenol	ug/kg	ND	146000	133000	144000	165000	99	125	17-107	14	20	M0
2-Methylnaphthalene	ug/kg	ND	146000	133000	159000	171000	109	129	20-120	7	20	M1
4-Chloro-3-methylphenol	ug/kg	ND	146000	133000	185000	176000	127	133	22-117	5	20	M1
4-Nitrophenol	ug/kg	ND	146000	133000	<350000	<318000	111	138	10-131		20	M0
Acenaphthene	ug/kg	ND	146000	133000	161000	169000	110	128	19-115	5	20	M0
Acenaphthylene	ug/kg	ND	146000	133000	174000	167000	119	126	11-122	4	20	M0
Anthracene	ug/kg	ND	146000	133000	181000	171000	124	129	10-133	6	20	
Benzo(a)anthracene	ug/kg	ND	146000	133000	196000	188000	134	142	10-137	4	20	M0
Benzo(a)pyrene	ug/kg	ND	146000	133000	184000	172000	126	130	10-132	7	20	
Benzo(b)fluoranthene	ug/kg	ND	146000	133000	197000	182000	136	137	10-134	8	20	M0
Benzo(g,h,i)perylene	ug/kg	ND	146000	133000	178000	166000	122	126	10-138	7	20	
Benzo(k)fluoranthene	ug/kg	ND	146000	133000	196000	183000	135	138	10-140	7	20	
Chrysene	ug/kg	ND	146000	133000	188000	174000	129	132	10-150	7	20	
Dibenz(a,h)anthracene	ug/kg	ND	146000	133000	185000	172000	127	130	10-126	7	20	M1
Fluoranthene	ug/kg	ND	146000	133000	195000	183000	134	138	10-128	6	20	M1
Fluorene	ug/kg	ND	146000	133000	171000	168000	117	127	12-127	2	20	
Indeno(1,2,3-cd)pyrene	ug/kg	ND	146000	133000	185000	170000	127	129	10-125	8	20	M1
N-Nitroso-di-n-propylamine	ug/kg	ND	146000	133000	147000	157000	101	118	23-101	6	20	M1
Naphthalene	ug/kg	ND	146000	133000	151000	163000	104	123	12-120	8	20	M0
Pentachlorophenol	ug/kg	ND	146000	133000	<350000	<318000	90	122	10-112		20	M1
Phenanthrene	ug/kg	ND	146000	133000	184000	169000	126	128	10-125	8	20	M0
Phenol	ug/kg	ND	146000	133000	156000	163000	107	123	22-105	4	20	M0, P1

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2814568 2814569												
Parameter	Units	5263625005		MS	MSD	MS	MSD	MS	MSD	% Rec	Limits	Max
		Result	Spike	Conc.	Conc.							
Pyrene	ug/kg	ND	146000	133000	208000	188000	143	142	10-146	10	20	
2,4,6-Tribromophenol (S)	%.						59	64	20-109			
2-Fluorobiphenyl (S)	%.						42	48	36-91			
2-Fluorophenol (S)	%.						43	51	24-95			
Nitrobenzene-d5 (S)	%.						41	48	26-96			
p-Terphenyl-d14 (S)	%.						68	70	27-117			
Phenol-d5 (S)	%.						43	52	32-93			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 610315

Analysis Method: EPA 8270

QC Batch Method: EPA 3510

Analysis Description: 8270 TCLP MSSV

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2813108

Matrix: Water

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	03/16/21 14:24	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	03/16/21 14:24	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	03/16/21 14:24	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	03/16/21 14:24	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	03/16/21 14:24	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	03/16/21 14:24	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	03/16/21 14:24	
Hexachlorobenzene	mg/L	<0.010	0.010	03/16/21 14:24	
Hexachloroethane	mg/L	<0.010	0.010	03/16/21 14:24	
Nitrobenzene	mg/L	<0.010	0.010	03/16/21 14:24	
Pentachlorophenol	mg/L	<0.050	0.050	03/16/21 14:24	
Pyridine	mg/L	<0.010	0.010	03/16/21 14:24	
2,4,6-Tribromophenol (S)	%	99	54-106	03/16/21 14:24	
2-Fluorobiphenyl (S)	%	87	42-95	03/16/21 14:24	
2-Fluorophenol (S)	%	55	23-70	03/16/21 14:24	
Nitrobenzene-d5 (S)	%	88	49-100	03/16/21 14:24	
p-Terphenyl-d14 (S)	%	119	52-124	03/16/21 14:24	
Phenol-d5 (S)	%	33	10-51	03/16/21 14:24	

LABORATORY CONTROL SAMPLE: 2813109

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.036	72	18-89	
2,4,5-Trichlorophenol	mg/L	0.05	<0.050	93	57-113	
2,4,6-Trichlorophenol	mg/L	0.05	0.046	91	63-105	
2,4-Dinitrotoluene	mg/L	0.05	0.045	89	48-112	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.038	76	46-93	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.070	70	40-88	
Hexachloro-1,3-butadiene	mg/L	0.05	0.035	71	10-94	
Hexachlorobenzene	mg/L	0.05	0.039	78	46-75 L1	
Hexachloroethane	mg/L	0.05	0.035	70	10-87	
Nitrobenzene	mg/L	0.05	0.044	87	51-108	
Pentachlorophenol	mg/L	0.05	<0.050	67	21-115	
Pyridine	mg/L	0.05	0.026	51	10-78	
2,4,6-Tribromophenol (S)	%			97	54-106	
2-Fluorobiphenyl (S)	%			85	42-95	
2-Fluorophenol (S)	%			55	23-70	
Nitrobenzene-d5 (S)	%			86	49-100	
p-Terphenyl-d14 (S)	%			107	52-124	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

LABORATORY CONTROL SAMPLE: 2813109

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			33	10-51	

MATRIX SPIKE SAMPLE: 2813110

Parameter	Units	50281495001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.32	63	19-91	
2,4,5-Trichlorophenol	mg/L	ND	0.5	<0.50	93	40-123	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.44	87	47-113	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.42	83	40-112	
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.36	72	38-97	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.66	66	36-91	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.32	64	11-98	
Hexachlorobenzene	mg/L	ND	0.5	0.37	73	38-77	
Hexachloroethane	mg/L	ND	0.5	0.31	63	10-93	
Nitrobenzene	mg/L	ND	0.5	0.38	77	44-106	
Pentachlorophenol	mg/L	ND	0.5	<0.50	66	22-126	
Pyridine	mg/L	ND	0.5	0.25	50	10-83	
2,4,6-Tribromophenol (S)	%.				95	54-106	
2-Fluorobiphenyl (S)	%.				76	42-95	
2-Fluorophenol (S)	%.				49	23-70	
Nitrobenzene-d5 (S)	%.				75	49-100	
p-Terphenyl-d14 (S)	%.				108	52-124	
Phenol-d5 (S)	%.				32	10-51	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch:	609245	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	Dry Weight/Percent Moisture
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

SAMPLE DUPLICATE: 2807724

Parameter	Units	50281333001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	96.8	96.8	0	5	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609243

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

SAMPLE DUPLICATE: 2807717

Parameter	Units	50281333001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	3.2	3.2	2	10	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609242

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

SAMPLE DUPLICATE: 2807715

Parameter	Units	50281333001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	56.2	56.1	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch: 609645	Analysis Method: EPA 9038
QC Batch Method: EPA 9038	Analysis Description: 9038 Sulfate Soil
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2809686 Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<100	100	03/11/21 13:53	N2

LABORATORY CONTROL SAMPLE: 2809687

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	199	99	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2809688 2809689

Parameter	Units	50281094001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	<955	1910	1910	2900	2840	106	103	90-110	2	20	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 608974

Analysis Method: EPA 9045

QC Batch Method: EPA 9045

Analysis Description: 9045 pH

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

SAMPLE DUPLICATE: 2806814

Parameter	Units	50280655001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	3.0	3.0	0	2	H3,PO

SAMPLE DUPLICATE: 2806815

Parameter	Units	50281295001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.1	8.1	0	2	H3

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch:	379755	Analysis Method:	EPA 351.2
QC Batch Method:	EPA 351.2	Analysis Description:	351.2 TKN
		Laboratory:	Pace Analytical Services - Green Bay

Associated Lab Samples: 50281333001

METHOD BLANK: 2190256 Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<100	100	03/15/21 18:52	

LABORATORY CONTROL SAMPLE: 2190257

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	500	478	96	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2190258 2190259

Parameter	Units	50281333001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	63300	14800	14600	77500	76900	96	93	80-120	1	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50281333

QC Batch: 609208

Analysis Method: EPA 353.2

QC Batch Method: EPA 353.2

Analysis Description: 353.2 Nitrate + Nitrite

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2807623

Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	03/09/21 11:47	B0,N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	03/09/21 11:47	B0,N2

LABORATORY CONTROL SAMPLE: 2807624

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	9.9	99	90-110	B0,N2
Nitrogen, NO2 plus NO3	mg/kg	20	19.6	98	90-110	B0,N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2807625 2807626

Parameter	Units	50281440006 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	ND	10.9	10.9	16.0	16.3	100	103	90-110	1	20	B0,N2
Nitrogen, NO2 plus NO3	mg/kg	ND	21.6	21.6	26.5	26.8	98	100	90-110	1	20	B0,N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch:	609628	Analysis Method:	SM 4500-NH3 G
QC Batch Method:	SM 4500-NH3 B	Analysis Description:	4500 Ammonia, Distilled
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2809647 Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	03/11/21 14:04	N2

LABORATORY CONTROL SAMPLE: 2809648

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.8	21.5	109	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2809649 2809650

Parameter	Units	50281333001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	27900	627	627	28500	27800	88	-24	90-110	2	20	N2,P6

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50281333

QC Batch: 609882	Analysis Method: SM 4500-Cl-E
QC Batch Method: SM 4500-Cl-E	Analysis Description: 4500 Chloride
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50281333001

METHOD BLANK: 2810886 Matrix: Solid

Associated Lab Samples: 50281333001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.4	99.4	03/12/21 08:12	N2

LABORATORY CONTROL SAMPLE: 2810887

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	199	100	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2810888 2810889

Parameter	Units	50281094001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	<953	1900	1900	2620	2630	103	103	90-110	0	20	N2

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50281333

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

BATCH QUALIFIERS

Batch: 609250

[B0] Analyte was detected in an associated blank at a concentration greater than the MDL.

ANALYTE QUALIFIERS

1d Sample was analyzed at a high dilution due to matrix being oil. TMW 03-17-21
B0 Analyte was detected in an associated blank at a concentration greater than the MDL.
E Analyte concentration exceeded the calibration range. The reported result is estimated.
H3 Sample was received or analysis requested beyond the recognized method holding time.
L1 Analyte recovery in the laboratory control sample (LCS) was above QC limits. Results for this analyte in associated samples may be biased high.
M0 Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1 Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M3 Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2 The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1 Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
P6 Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
PO The reported result is outside the range of the pH buffer solutions used to check the calibration of the pH meter.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50281333

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50281333001	Digester #4	EPA 3546	609930	EPA 8081	610183
50281333001	Digester #4	EPA 3510	610318	EPA 8081	610410
50281333001	Digester #4	EPA 3546	610635	EPA 8082	610719
50281333001	Digester #4	EPA 8151	610667	EPA 8151A	610789
50281333001	Digester #4	EPA 3050	609624	EPA 6010	610151
50281333001	Digester #4	EPA 3010	609722	EPA 6010	610094
50281333001	Digester #4	EPA 3050B	609244	EPA 6020	609515
50281333001	Digester #4	EPA 7470	609918	EPA 7470	610144
50281333001	Digester #4	EPA 7471	609472	EPA 7471	609619
50281333001	Digester #4	EPA 3546	610663	EPA 8270	610773
50281333001	Digester #4	EPA 3510	610315	EPA 8270	610439
50281333001	Digester #4	EPA 8260	610552		
50281333001	Digester #4	EPA 5030/8260	610542		
50281333001	Digester #4	SM 2540G	609245		
50281333001	Digester #4	SM 2540G	609243		
50281333001	Digester #4	SM 2540G	609242		
50281333001	Digester #4	EPA 9038	609645	EPA 9038	609672
50281333001	Digester #4	EPA 9045	608974		
50281333001	Digester #4	EPA 351.2	379755	EPA 351.2	379797
50281333001	Digester #4	EPA 353.2	609208	EPA 353.2	609250
50281333001	Digester #4	SM 4500-NH3 B	609628	SM 4500-NH3 G	609754
50281333001	Digester #4	SM 4500-CI-E	609882	SM 4500-CI-E	609899

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Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>3/4/21 1500</u>		Evaluated by: <u>MEE</u>		WO#: 50281333 PM: MSB Due Date: 03/18/21 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>					
Project Manager: <u>MSB</u>		Profile ID:			
Rush TAT Requested: YES <u>NO</u> Due Date:					
Lab Notified of Rush or Short Holds: YES NO				Non Conformance Form Required: YES <u>NO</u>	
Samples Received Via: FedEx <u>UPS</u> Client Pace Courier Other: _____					Comments:
Custody Seals Present and Intact:		<u>YES</u>	NO	NA	
Received Sample Information Form(s): Drinking Waters Only		YES	NO	<u>NA</u>	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES	<u>NO</u>	N/A	
Short Holds Present (< 72 Hours):		YES	<u>NO</u>		
Samples Received in Hold:		<u>YES</u>	NO		
Custody Signatures Present:		<u>YES</u>	NO		
Collector Signature Present:		<u>YES</u>	NO		
Packing Material Used:		<u>YES</u>	NO		
Samples Collected Today and On Ice:		YES	NO	<u>N/A</u>	
IR Gun #: <u>280</u> 281			Digital Thermometer #: 282 283		
Ice Type: WET Bagged / <u>WET Loose</u> BLUE NONE			1. Cooler Temp Upon Receipt: <u>4.4 / 4.9</u> °C		
Ice Location: <u>TOP</u> BOTTOM MIDDLE DISPERSED			Temp should be 0-6°C (Initial/Corrected)		
Temp Blank Received:		YES	<u>NO</u>		
Containers Intact:		<u>YES</u>	NO		
Correct Containers:		<u>YES</u>	NO		
Sufficient Volume:		<u>YES</u>	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: _____ <i>Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl</i>		YES	NO	<u>N/A</u>	
Residual Chlorine Absent: Cl₂ Strip Lot #: _____ (SVOC/Pest 625, PCB 608, Total/Amenable/Available Cyanide)		YES	NO	<u>N/A</u>	
VOA Headspace Acceptable (<6mm):		YES	NO	<u>N/A</u>	
Trip Blank Received: HCl MeOH TSP OTHER		YES	<u>NO</u>		
Comments:		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			